A set symmetric for a sparse statistic is Schur-positive with an explicit two-row expansion

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Editorial note. The counted statement is the author's labelled lemma 'lem:if two rows and symmetric then schur positive' (source lines 612-618); the proof environment begins on the next line after \end{lemma}, so the pairing is by adjacency.

Editorial note. Three context spans are retained: the author's Schur-positivity criterion, and the two superstandard characterisations that the proof invokes; every label used inside the retained text is therefore defined inside the package.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source; packages used only elsewhere in the article are left out because no retained line needs them.

Editorial note. No retained line refers to a label, citation or figure outside the retained spans, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: Schur-positivity of a set symmetric for a sparse statistic with its two-row expansion (author label lem:if two rows and symmetric then schur positive).

Allowed standard prerequisites (editorial).

- Symmetric set with respect to a statistic. A is symmetric for D : A -> 2^{[N-1]} if the generating function sum over a in A of t^{D(a)} expands in standard Young tableaux with coefficients depending only on the shape.. It is the hypothesis that produces equation (1) at source lines 620-623.
- Sparse statistic. D(a) is sparse if it contains no two consecutive integers.. It is used at source line 634 to rule out shapes with more than two parts.
- Superstandard tableau characterisations. The descent set of T_lambda determines the dominance order of the shape, and in the two-row case it determines the tableau and the shape uniquely.. Lemma 1 at source lines 632-634 and Lemma 2 at source line 641 are the two inputs that turn the generating-function identity into the coefficient formula.

Source reference. Original theorem 2.5: sources/alco-351-matching-schur-2024/publisher-source.tex, source lines 368–368. Original criterion of Adin and Roichman cited by the retained proof at source lines 620 and 632. It is a navigation pointer: the counted lemma's own proof is retained line for line and nothing is replaced by this reference. The reference number was inferred by counting the theorem-class environments of that section of the pinned source; it is not confirmed by the retained bibliography or by a label-number dump of the original, so it must be read as a navigation pointer only.

The power of these notions may be reflected by the following statement:

Lemma 0.1. Let $\lambda, \mu \vdash N$ be two partitions. Then:

1. $\mathrm{Des}(T_\lambda) = [N-1] \setminus \{\lambda'_1, \lambda'_1 + \lambda'_2, \dots\}$.
2. If $\mathrm{Des}(T) = [N-1] \setminus \{\lambda'_1, \lambda'_1 + \lambda'_2, \dots\}$ for some $T \in \mathrm{SYT}(\mu)$, then $\mu' \geq \lambda'$. Furthermore, if $\mu = \lambda$ then $T = T_\lambda$.

Lemma 0.1 associates the Young diagram of shape λ with the set $\mathrm{Des}(T_\lambda)$. As we will see later, this association is powerful in analysing the Schur coefficients of symmetric sets.

Standard Young tableaux with at most 2 rows have a slightly stronger property:

Lemma 0.2. Let $\lambda = (N-k_1, k_1)$ and $\mu = (N-k_2, k_2)$ be two partitions of N with at most two parts each. Then:

1. $\mathrm{Des}(T_\lambda) = \mathrm{odds}(k_1)$.
2. If $\mathrm{Des}(T) = \mathrm{odds}(k_1)$ for some $T \in \mathrm{SYT}(\mu)$, then $k_1 = k_2$ and $T = T_\lambda$.

Lemma 0.3. Let $\mathcal{A}$ be a symmetric set with respect to a sparse statistic $D : \mathcal{A} \to 2^{[N-1]}$, and denote $n = \lfloor \frac{N}{2} \rfloor$. Then $\mathcal{A}$ is Schur-positive, and its Schur expansion is

$$\mathcal{Q}_D(\mathcal{A}) = \sum_{k=0}^{n} |\mathcal{A}(D = \mathrm{odds}(k))| \; s_{N-k,k}.$$

Proof. The set $\mathcal{A}$ is assumed to be symmetric, so by Theorem 2.5, we have

$$\sum_{a \in \mathcal{A}} \boldsymbol{t}^{D(a)} = \sum_{\lambda \vdash N} c_\lambda \sum_{T \in \mathrm{SYT}(\lambda)} \boldsymbol{t}^{\mathrm{Des}(T)}, \tag{0.1}$$

where c_λ are the Schur coefficients of $\mathcal{A}$. It suffices to show that if there exists k such that $\lambda = (N-k, k)$ then $c_\lambda = |\mathcal{A}(D = \mathrm{odds}(k))|$, and otherwise $c_\lambda = 0$.

If $\mathcal{A} = \emptyset$, then $\mathcal{Q}_D(\mathcal{A}) = 0$, and the statement holds. Therefore, we may assume that $\mathcal{A} \neq \emptyset$, and consequently, there exists $c_\lambda \neq 0$ for some partition $\lambda \vdash N$. Let $\mu \vdash N$ be a partition with $c_\mu \neq 0$, maximal in the conjugate order (i.e. such that $c_\lambda = 0$ for every $\lambda \vdash N$ with $\lambda' > \mu'$).

Equation (0.1) implies that the equation

$$|\mathcal{A}(D = J)| = \sum_{\lambda \vdash N} c_\lambda \, |\mathrm{SYT}(\lambda)(\mathrm{Des} = J)| \tag{0.2}$$

holds for all $J \subseteq [N-1]$. Let us find the right-hand side of Equation (0.2) when substituting $J = \mathrm{Des}(T_\mu)$. Let $T \in \mathrm{SYT}(\lambda)$ be a tableau with $\mathrm{Des}(T) = \mathrm{Des}(T_\mu)$. By Lemma 0.1, we have $T = T_\mu$ or $\lambda' > \mu'$. However, if $\lambda' > \mu'$ then $c_\lambda = 0$. Therefore, if $T \in \mathrm{SYT}(\lambda)$ has $\mathrm{Des}(T) = \mathrm{Des}(T_\mu)$ and $c_\lambda \neq 0$, then $T = T_\mu$. Consequently, substituting $J = \mathrm{Des}(T_\mu)$ into Equation (0.2), we find that $|\mathcal{A}(D = \mathrm{Des}(T_\mu))| = c_\mu \neq 0$.

We may conclude that there exists an element $a \in \mathcal{A}$ with $D(a) = \mathrm{Des}(T_\mu)$. The set $D(a)$ is sparse, so $\{1, 2\} \not\subseteq D(a)$. Lemma 0.1 implies that $\{1, 2\} \subseteq \mathrm{Des}(T_\mu)$ whenever $\mu'_1 > 2$, so we may deduce that $\mu'_1 \leq 2$. Therefore, $c_\lambda = 0$ for every partition λ with more than 2 parts.

Thus, we can reformulate Equation (0.2) as

$$|\mathcal{A}(D = J)| = \sum_{k=0}^{n} c_k \, |\mathrm{SYT}(N-k, k)(\mathrm{Des} = J)| \, . \tag{0.3}$$

By Lemma 0.2, since only partitions into at most two parts are involved in Equation (0.3), substituting $J = \mathrm{Des}(T_\lambda)$ for $\lambda = (N-k, k)$ yields $|\mathcal{A}(D = \mathrm{odds}(k))| = c_k$, as required. □